pset1

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1 18.335 Problem Set 1

Simon Opsahl (sopsahl@mit.edu)

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2 Problem 0: Pset Honor Code

I will not look at 18.335 pset solutions from previous semesters. I may discuss problems with my classmates or others, but I will write up my solutions on my own.

3 Problem 1: Floating point

Non-zero floating point numbers $\mathbb{F}$ can be written as

$$x = \pm(\frac{m}{\beta^t})\beta^e$$

with $\beta \geq 2$ and $\beta^{-1} \leq \frac{m}{\beta^t} < 1$. e is an unrestricted integer.

3.1 part (a)

In order to identify the smallest postive integer n that is not represented by $\mathbb{F}$, we must first identify that for integers $\in [0, \beta^t)$, we have a direct mapping of the mantissa to integer value because there are t digits of precision. For the value β^t, we can trivially represent it as a power of β. For the case of $\beta^t + 1$, however, we must have the assumed digit as 1, and then t additional digits to represent the 1 offset. As this requires $t + 1$ digits of precision, but our mantissa has a precision of t, $\beta^t + 1$ is the smallest positive integer not in $\mathbb{F}$.

For fp32, $t = 24$ and thus this value is $2^{24} + 1$. The result can be verified by the code below.

[1]:
```
beta = big(2.0)
t = big(24)

smallest_unsupported_value = Float32(beta^t + 1)
one_below = Float32(beta^t)
two_below = Float32(beta^t - 1)
smallest_unsupported_value == one_below, one_below==two_below
```

```
(true, false)
```

For fp64, $t = 53$ and thus this value is $2^{53} + 1$. The result can be verified by the code below.

[3]:
```
beta = big(2.0)
t = big(53)

smallest_unsupported_value = Float64(beta^t + 1)
one_below = Float64(beta^t)
two_below = Float64(beta^t - 1)
smallest_unsupported_value == one_below, one_below==two_below
```

```
(true, false)
```

3.2 part (b)

In base 2, we can prove that for two floating point numbers $a \leq b$, the following inequality holds:

$$a \leq fl(\frac{a+b}{2}) \leq b$$

This proof relies on the fact that $a \leq b \implies fl(a) \leq fl(b)$. We can first construct the following:

$$a \leq b$$

$$a + b \leq b + b = 2 \cdot b$$
$$fl(a+b) \leq fl(2 \cdot b)$$
$$\frac{fl(a+b)}{2} \leq \frac{fl(2 \cdot b)}{2}$$
$$fl(\frac{fl(a+b)}{2}) \leq fl(\frac{fl(2 \cdot b)}{2})$$

Because we are doing operations in base-2, the effect of mutiplying by 2 is increasing the exponent by 1, and the effect of dividing is decreasing the exponent by 1. As a result, $fl(\frac{fl(2 \cdot b)}{2}) = b$. The same logic can be used to prove that $a \leq fl(\frac{a+b}{2})$. As a result, in base-2, the following is always true if $a \leq b$:

$$a \leq fl(\frac{a+b}{2}) \leq b$$

[147]:
```
a = 6.1111
b = 6.1112

round(round(a+b, sigdigits=5)/2, sigdigits =5) >= round(a, sigdigits=5)
```

```
false
```

4 Problem 3: Matrix norm

4.1 part (a)

We aim to prove that $\|A\|_\infty = \max\limits_{1\le i\le m} \|a_i^*\|_1$. In order to do so, we can use the property of subordinate norms described in lecture that given $x \in \mathbb{C}^n$, we have $\|A\|_\beta = \max\limits_{\|x\|_\beta=1} \|Ax\|_\beta$. In this problem, $\beta = \infty$, and thus we find

$$\|A\|_\infty = \max_{\|x\|_\infty=1} \|Ax\|_\infty = \max_{\|x\|_\infty=1} (\max_{1\le i\le m} |a_i^* x|)$$

We also know that $\|x\|_\infty = \max\limits_{1\le j\le n} |x_j|$, and thus $|x_j| \le 1$. As a result, we can rewrite our norm as

$$\|Ax\|_\infty = \max_{1\le i\le m} |a_i^* x| \le \max_{1\le j\le n} \sum_{j=1}^{n} |a_{i,j}||x_j| \le \max_{1\le j\le n} \sum_{j=1}^{n} |a_{i,j}| = \max_{1\le j\le n} \|a_i^*\|_1$$

By choosing $x = sign(a_i^*)$, we attain this bound, and thus $\|A\|_\infty = \max\limits_{1\le i\le m} \|a_i^*\|_1$.

[14]:
```
using LinearAlgebra

# Generate a random A
m = rand(1:100)
n = rand(1:100)
A = rand(m, n)

# Calculate the row_sums
row_sums = sum(abs.(A), dims=2)

maximum(row_sums) == opnorm(A, Inf)
```

```
true
```

4.2 part (b)

The proof that $I + B$ is nonsingular relies on the lemma from lecture that the spectral radius is bounded by the norm. Thus, for $1 \le k \le n$, $|\lambda_k| \le \|B\| < 1$, and thus $|\lambda_k + 1| \ne 0$. As a result, the corresponding matrix is nonsingular.

In order to identify that $\|(I + B)^{-1}\| \le \frac{1}{1-\|B\|}$, we must first remark that B is small. Thus, we can expand the inversion with the Neumann series.

$$(I + B)^{-1} = \sum_{k=0}^{\infty} (-B)^k$$

If we take the norm of each side, then we get

$$\|(I + B)^{-1}\| = \|\sum_{k=0}^{\infty} (-B)^k\| \le \sum_{k=0}^{\infty} \|(-B)^k\| \le \sum_{k=0}^{\infty} \|B\|^k = \frac{1}{1 - \|B\|}$$

5 Problem 4: SVD

5.1 part (a)

In this problem, we must find the 2-norm and 2-condition number of the nonsingular matrix

$$C = \begin{bmatrix} \alpha I & A^* \\ A & \alpha I \end{bmatrix}$$

given that $A \in \mathbb{C}^{nxn}$ with singular values $\sigma_1 \geq ... \geq \sigma_n > 0$. From lecture, we know that $\|C\|_2$ is equivalent to the largest singular value of C, and $\kappa_2(C)$ is the fraction of the largest singular value over the smallest.

As a result, all that is left is to find the singular values of C. They can be found from the square root of the eigenvalues of C^*C.

$$M = C^*C = \begin{bmatrix} \alpha^2 I + A^*A & 2\alpha A^* \\ 2\alpha A & \alpha^2 I + AA^* \end{bmatrix}$$

As the eigenvalues of A^*A are equivalent to Σ^2, then we can compute the eigenvalues of M with the following formula:

$$\lambda_i^{+/-} = \frac{2\alpha^2 + 2\sigma_i^2 \pm \sqrt{(2\alpha^2 + 2\sigma_i^2)^2 - 4(\alpha^4 + \sigma_i^4 - 2\alpha^2\sigma_i^2)}}{2} = \alpha^2 + \sigma_i^2 \pm 2\alpha\sigma_i = (\alpha \pm \sigma_i)^2$$

The singular values of C are the square root of the eigenvalues of M, and thus we have

$$\mu_i^{+/-} = |\alpha \pm \sigma_i|$$

[82]:
```
using LinearAlgebra

n = 5
A = rand(n, n)*10

alpha = rand()*10
while any(abs.(svd(A).S .- alpha) .< 1e-10)
    alpha = rand()*10
end



I_ = I(n)

largest = maximum(vcat(alpha .+ svd(A).S, abs.(alpha .- svd(A).S)))
smallest = minimum(vcat(alpha .+ svd(A).S, abs.(alpha .- svd(A).S)))
largest, largest/smallest
```

```
(39.7237754680893, 28.446865343671117)
```

[83]:
```
I_ = I(n)
C = [alpha*I_ A'; A alpha*I_]

opnorm(C), cond(C)
```

```
(39.72377546808932, 28.446865343671174)
```

5.2 part (b)

In order to identify that $1 \leq \text{srank}(A) \leq \text{rank}(A)$, we must first expand both the frobenius norm and the 2-norm.

$$\|A\|_F^2 = tr(A^*A) = \sum_{i=1}^{r} \sigma_i^2$$

In addition,

$$\|A\|_2^2 = \sigma_1^2$$

Because $\forall i \in (1, r], \sigma_i \leq \sigma_1$, and there are r singular values because $\text{rank}(A^*A) = \text{rank}(A)$, we know that $\sum_{i=1}^{r} \sigma_i^2 \leq r \cdot \sigma_1^2$. As a result,

$$\text{srank}(A) = \frac{\sum_{i=1}^{r} \sigma_i^2}{\sigma_1^2} \leq r = \text{rank}(A)$$

In addition, we know trivially that $\sum_{i=1}^{r} \sigma_i^2 \geq \sigma_1^2$, and thus

$$\text{srank}(A) = \frac{\sum_{i=1}^{r} \sigma_i^2}{\sigma_1^2} \geq 1$$

[40]:
```
using LinearAlgebra

function srank(A)
    A_F =  norm(A)
    A_2 = opnorm(A, 2)
    return A_F^2 / A_2^2
end
```

```
srank (generic function with 1 method)
```

[44]:
```
m = rand(1:100)
n = rand(1:100)
A = rand(m, n)

1 <= srank(A) <= rank(A)
```

```
true
```